

Metropolitan University

Department of Computer Science and Engineering

Spring Term Final Examination – 2025

Program: B.Sc. in CSE; Batch: 58

Course: GED-215: Industrial Management & Financial Accounting

Time: 2 Hours

Answer any **four** sets of questions. All parts of each question must be answered sequentially.
Marks: 40
Figures in the right margin indicate full marks.

- 1) Selected transactions for Lilly Company during the first month in business are presented below:

- July 1 Invested Tk.30,000 cash in the business.
5 Purchased equipment for Tk. 12,000, paying Tk.6, 000 in cash and the balance on account.
20 Paid Prepaid insurance Tk.3, 000
22 Withdrew Tk.1500 cash for office use.
25 Paid Tk.3, 000 cash on balance owed for equipment purchased.
30 Withdrew Tk.500 cash for personal use.

Requirement:

- a) Prepare journal entries and post to the ledger account. 6
b) Prepare a trial balance. 4
- 2) a) What is accrued revenue? Give an example. 2
b) The following is the Trial Balance of Orni Enterprise for the quarter ended March 31, 2021.

Accounts Title	Debit (Tk.)	Credit (Tk.)
Cash	180,400	
Accounts receivable	145,200	
Prepaid insurance	46,800	
Equipment	624,000	
Mortgage payable		947,800
Notes payable		123,000
Orni capital		1,150,000
Service revenue		141,800
Land	640,000	
Insurance expense	7,800	
Building	1,288,000	
Depreciation expense	73,600	
Accumulated Depreciation-Building		456,000
Interest expense	26,000	
Interest payable		26,000
Accumulated Depreciation- Equipment		187,200
Total	30,31,800	30,31,800

Others data:

- Depreciation of Tk. 5,000 per quarter on equipment.
- Interest accrued on mortgage payable, issued on January 1, Tk. 3,000.

3. Insurance expires at the rate of Tk. 1,500 per month,
4. Service provided on account at March 31, Tk. 7,500.
5. Salaries accrued but not paid Tk. 1000.
6. 10% interest on notes payable accrued but not paid.

Required: Prepare Adjusting Entries

8

3. By using the data of above Question no. 2(b)

Required:

- a) Prepare income statement
 - b) Prepare a classified balance sheet assuming the notes payable is short-term. 5
5
4. (a) What is planning? Why do managers plan? 3
(b) Differentiate between strategic and operational planning. 3
(c) Define MBO and Traditional objective setting. Give an example of traditional goal setting pyramid. 4
 5. (a) 'To motivate someone, you need to understand what need level that person is on in the hierarchy'. Based on this statement, how does Maslow's theory explain motivation? 4
(b) Briefly explain Equity Theory and what happens if there are inequities at work. 3
(c) Define Goal Setting Theory and show its details in a diagram. 3
 6. (a) "Leaders are born not made" do you agree? Describe the contribution of University of Iowa in Behavioral theories of leadership. 5
(b) Do you think managerial grid is the alternative of behavioral theories provided by Ohio State University and University of Michigan? Provide appropriate arguments regarding your judgments. 5

Metropolitan University

Department of Computer Science and Engineering

Spring Term Final Examination – 2025

Program: B.Sc. in CSE; Batch: 58(All Sections)

Course: MAT 235: Numerical Methods

Time: 2 Hours

Marks: 40

Answer any four questions. All parts of each question must be answered sequentially.

Figures in the right margin indicate full marks.

1. a) Explain Newton Raphson formula from Geometry. 3
b) Find the real root of the equation $x^3 - 2x - 5 = 0$ using bisection method correct to 2 decimal places. 7

2. a) Solve the following system using LU factorization: $\begin{cases} 3x + 2y + z = 1 \\ 2x + 3y + 2z = 2 \\ 5x + 4y + 3z = 3 \end{cases}$ 7

- b) Use the Gauss-Jordan method to solve the system of linear equations: $\begin{cases} x - 2y + 3z = 5 \\ 2x - 4y + 6z = 10 \\ 3x - 6y + 9z = 15 \end{cases}$ 3

3. a) Discuss Iterative Method. 2
b) Find a root of the equation $4e^x \sin x - 1 = 0$ correct to 3 decimal places by regula-falsi method given that the root lies between 0 and 0.5. 8

4. a) Using the table given below find $f(x)$ as a polynomial in x by using Newton's divided difference formula. 6
b) Find the output of $x = 5$ for the following data set using Lagrange Polynomial Interpolation. 4

x	-1	0	3	6	7
$f(x)$	3	-6	39	822	1611

5. a) Derive the formulas for the first and second derivatives using Newton's Forward Difference Interpolation Formula. 5
b) Find the first and second derivatives of the function given below at the point $x = 3.5$ 5

x	1	2	3	4	5
y	0	1	5	6	8

6. $\int_{0.2}^{1.4} (\cos x - \ln x + e^{2x}) dx$
a) Compute the approximate value of the integral using each of the methods: i) Trapezoidal Rule ii) Simpson's 1/3 Rule iii) Simpson's 3/8 Rule iv) Weddle's Rule 6
b) Determine the true value of the integral analytically. 2
c) Compare the accuracy of the methods based on the absolute errors. 2

Metropolitan University
Department of Computer Science and Engineering
Spring Term Final Examination – 2025
Program: B.Sc. in CSE; Batch: 58
Course: CSE 221: Object Oriented Programming

Marks: 40

Time: 2 Hours

Answer any **four** questions. All parts of each question must be answered sequentially.
Figures in the right margin indicate full marks.

1.
 - a) Write a note about the principles of object-oriented programming. 4
 - b) Describe JVM. Discuss the main difference between machine code and bytecode. 2
 - c) Explain the differences in the compilation process, using figures, between **Structured Programming** and **Object-Oriented Programming**. 4
2.
 - a) What is the relationship between a class and an object in Java, and how do they contribute to Object-Oriented Programming (OOP)? 4
 - b)
 - I. Create a constructor for a class named **Student** with 2 parameters and set the instance variables **rollNumber** and **studentName** inside it. 6
 - II. Create a method **showInfo()** and overload the method with a String parameter and print the string inside it.
 - III. Write the statement to create an object of class **Student** using the constructor of question number 2.b(I). 6

3. a) Rewrite the code with finding bugs(errors) and solve these and show the output. 6

```
public class Phone {
    int battery;
    phone(int battery){
        this.battery = battery;
    }
    void call() {
        System.out.println(battery+"
mAh");
        System.out.println("Calling");
    }
}
```

```
public class SmartPhone extend Phone{
    double camera;
    SmartPhone(double camera, int
battery){ super.battery = battery;
        this.camera = camera;
    }
    void selfie()
        super.call();
        System.out.println(camera+"
MP");
        System.out.println("Taking
Selfie");
    }
}
```

```
public class Main {
    public static void main(String[] args) {
        Phone p = new Phone(4000);
        p.call();
        SmartPhone sp = new SmartPhone(13);
        sp.selfie();
    }
}
```

- b) What are the differences between **Method Overloading** and **Method Overriding**?

4. a) Provide a code example of using a **try-catch** block in the handling of an exception.
b) Briefly describe the thread lifecycle in java.
c) What is multithreading? What benefits do we get from using it?
5. a) Explain the ways in which the available access modifiers in Java influence inheritance.
b) How would you implement a getter and setter method for a private integer variable **age** in a Java class? Include the code in your answer.
c) What is the difference between a **built-in package** and a **user-defined package** in Java?
d) In what scenarios should you prefer an abstract class over an interface in Java?
6. a) A method declared as final can be overridden in subclasses. True or False?
b) A _____ variable can be used without creating the instance of the class.
c) A class can have more than one constructor with different names. True or False?
d) Which keyword is used to create an object of a class?
e) What is the default value of a String type instance variable?
f) Declare an Integer array of size 100.
g) What is the return type of a constructor?
h) A constructor must be called by an object. True or False?
i) Static methods must be called using an object of the class. True or False?
j) When do we need to use the keyword **extends**?

Metropolitan University
Department of Computer Science and Engineering
Spring Term Final Examination – 2025
Program: B.Sc. in CSE; Batch: 58
Course: CSE 213: Computer Organization and Architecture

Time: 2 Hours

Total Marks: 40

*[Answer any **four** sets of questions. All parts of each question must be answered sequentially. Figures in the right margin indicate full marks.]*

1. (a) What makes Programmed I/O different from Interrupt Driven I/O in terms of how they use the CPU? (2)
(b) Draw the circuit diagram of a DMA with various types of devices and write its working principles. (5)
(c) Describe the different DMA transfer modes used in a computer system. (3)
2. (a) A CPU uses **Booth's** algorithm for signed binary multiplication. Using this method, compute the product of **-9** and **+4**. Show the Booth's table, all intermediate calculations, and the final result. (6)
(b) A CPU executes a program with the following parameters:
Instruction Count (IC) = 500,000 instructions
Clock Rate = 2.0 GHz
CPI (Cycles Per Instruction) = 6
(i) Calculate the total execution time of the program in seconds.
(ii) If the clock rate decreases by 20%, what is the new execution time?
(iii) If the CPI increases by 20%, what is the new execution time?
(iv) Which change, decreasing clock speed or increasing CPI, results in a greater increase in execution time? (4)
3. (a) Outline the operational principles of direct mapping, associative mapping, and set-associative mapping in cache systems, emphasizing their impact on cache hit rates. (6)
(b) Illustrate a 3*3 DRAM and label its component. Provide a brief explanation of each component. (4)
4. (a) Write short notes on:
(i) Virtual Memory (ii) Page Fault (iii) Demand Paging (iv) Thrashing (4)
(b) Explain different types of bus arbitration systems with the figure. (6)

5. (a) A magnetic disk has the following characteristics:
Number of platters: 3
Tracks per platter: 400
Sectors per track: 200
Bytes per sector: 512
Evaluate the total storage capacity of the disk in megabytes (MB). (2)
- (b) Define RAID. Discuss RAID Levels (RAID-0 to RAID-6). (5)
- (c) Compare the write-through and write-back cache policies in terms of how they handle data consistency and performance. (3)
6. (a) Explain the process of executing an instruction using the CPU and memory with the help of a diagram. (4)
- (b) $A[25] = A[8] + A[15]$.
Assume the base address register used for the array A is \$s2. Generate the MIPS assembly code for the given expression? Also, write its MIPS instructions. (5)
- (c) Differentiate between general purpose register and special purpose register? (1)